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Part 1: Research Git & Github

What is git?

1. What is a version control system (VCS)?

Version Control System is a system that helps detecting and recording the changes that have been made in file, project, code etc. There are *distributed* and *centralized* VCS systems. One example of distributed VCS is Git.

2. What is Git and what problem does it solve?

It's a distributed VCS that helps to record the changes in the file.

3. What is a repository (repo)?

It's a database that collects all the necessary information about the changes in the project.

4. What is a commit? Why do we write a commit message?

Commit is like a "save" button, but more powerful as it helps to return to every record that has been saved, available for everybody in the team. Basically, a commit message is needed to easily grasp the main point of the change, like a comment that tells you something was done here.

5. What is the difference between git add and git commit?

Git add adds the changes, *git commit* commits (confirms) them.

6. What does git push do? What does git pull do?

When change has been done, it needs to be shared with other members of the team. Primarily it's saved only on the local machine, but to ensure everybody can see it, we can use git push, so it's saved in a remote repository. Same but reversed is git pull.

7. What is a branch and why is it useful?

Branch is like a parallel space where we can conduct changes without affecting the main document, and if we are happy with the result, we can merge the branch back in. This is a useful thing, because it helps to check alternative ways.

What is GitHub?

8. What is GitHub and how is it different from Git?

GitHub is a website that connects repositories with external members, while Git only tracks changes locally.

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9. Name two Alternatives to GitHub?

GitLab and Bitbucket

10. What is the difference between a public and a private repository?

Public is for public audience, private is for specific organizations with sensitive information.

11. What is a *ReadME.md* file and what is it used for?

It's individual, however most of the ReadMe.md files are quite the same, it's like an instruction for usage of folder, agenda that tells what was done etc.

12. What is a *.gitignore* file and why would you not want to commit every file?

.Gitignore tells Git to never track or commit those files, the main reason would be security and sensitivity of the data provided there.

Used sources:

What is Git? Explained in 2 Minutes!

https://www.geeksforgeeks.org/git/version-control-systems/

What is Git and GitHub?

How Git Works: Explained in 4 Minutes